

Antipsychotic Drugs

Background

What is psychosis?

Psychosis refers to several symptoms that affect the mind. A person with psychosis will often experience disrupted thoughts and perceptions. It is difficult for the psychotic patient to determine what is and is not real. It usually starts in early adulthood, and may develop due to conditions such as Parkinson's disease or Alzheimer's disease and medications. Other common causes are mental health conditions, such as schizophrenia.

What is schizophrenia?

Schizophrenia is a mental health condition that affects thoughts, behaviors, and feelings. A person with schizophrenia may be presented with positive or negative symptoms.

The positive symptoms include hallucinations, delusions, illogical changes in behavior or thoughts, hyperactivity, and thought disorder.

The negative symptoms include apathy, lethargy, and withdrawal from social events or settings.

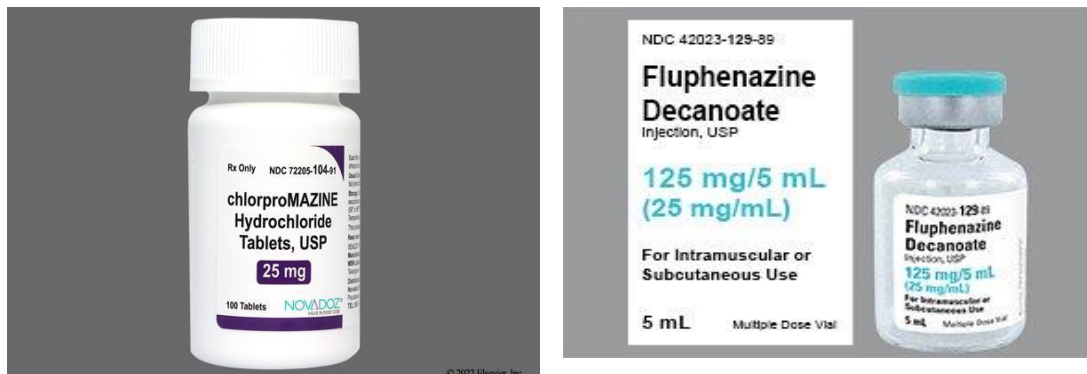
Antipsychotics, or neuroleptics agents are a major class of *drugs used to treat symptoms that happen with schizophrenia and* psychosis.

Classification of antipsychotic medications

A) **First-generation antipsychotics:** are dopamine receptor antagonists (DRA) and are known as **typical antipsychotics** (chlorpromazine, haloperidol, fluphenazine, pimozide,....

B) Second-generation antipsychotics : are, in general speaking, serotonin-dopamine antagonists and are also known as **atypical antipsychotics (clozapine, olanzapine Quetiopine, aripiprazole, risperidone,...)**

Examples of first-generation antipsychotics (typical) as chlorpromazine, *haloperidol*, fluphenazine.



Examples of second-generation antipsychotics (atypical) as clozapine , and risperidone.



A. Mechanism of action

1. Dopamine antagonism:

All of *1st generation* (typical, or traditional antipsychotics) and most of *2nd generation* block the post synaptic D2.

Bind tightly to D2 → ↑ Extrapyramidal symptoms (EPS).

Chlorpromazine bind weakly → Movement disorders are less

1st generation also have noradrenergic, cholinergic, and histaminergic blocking action.

2. Serotonin receptor–blocking activity: particularly 5-HT_{2A} receptors

Most of *2nd generation* (atypical antipsychotics) associated with a higher risk of metabolic side effect such as diabetes, hypercholesterolemia, and weight gain.

Clozapine: high affinity for D₁, D₄, 5-HT₂, M, and α-, a weak D₂

Risperidone & olanzapine: blocks 5-HT_{2A} > D₂ receptors,



aripiprazole: is a *partial agonist* at D₂ and 5-HT_{1A} receptors

Quetiapine: blocks D₂ more potently than 5-HT_{2A}.

Cariprazline, a new atypical antipsychotic, has been approved by the US Food and Drug Administration (FDA). It is a partial agonist of dopamine D₃ and 5-HT_{1A} receptors and has a good effect on schizophrenia and bipolar disorder.

B. Actions

The clinical effects depend on blockade at dopamine and/or serotonin receptors.

However, many of them also block cholinergic, adrenergic, and histaminergic → side effects

1. Antipsychotic effects:

All ↓ hallucinations & delusions of schizophrenia (positive symptoms)

The atypical (*clozapine*), can ↓ apathy, impaired attention, & cognitive impairment (negative symptoms)

2. Extrapyramidal effects:

- Blockade of dopamine receptors in the nigrostriatal pathway
- The second generation has lower incidence of EPS.
- Dystonia's (sustained contraction of muscles leading to twisting, distorted postures),
- Parkinson-like symptoms, akathisia (motor restlessness)
- Tardive dyskinesia (involuntary movements, usually of the tongue, lips, neck, trunk, and limbs)

3. Antiemetic effects:

Most of them have antiemetic by blocking D2 of the CTZ of the medulla

4. Anticholinergic effects:

Thioridazine, chlorpromazine, clozapine & olanzapine



Blurred vision, dry mouth (the exception is *clozapine*, which increases salivation), confusion, constipation and urinary retention.

The anticholinergic effects may reduce the risk of EPS with them.

5. Other effects:

- Blockade of α → orthostatic hypotension and light-headedness.
- Alter temperature-regulating mechanisms → poikilothermia (temperature varies with environment).
- In the pituitary, block D2 receptors, → ↑ prolactin release.
- Block H1-histamine, *chlorpromazine*, *olanzapine*, *quetiapine*, and *clozapine* → Sedation

C. Therapeutic uses

1. Treatment of schizophrenia:

The only efficacious pharmacological treatment for schizophrenia.

The 1st -generation are most effective in treating positive symptoms.

The atypical is effective in patients who are resistant & in treating the negative symptoms.

2. Prevention of nausea and vomiting:

The older antipsychotics *prochlorperazine* are useful in the treatment of drug-induced nausea.

3. Tranquilizers

To manage agitated and disruptive behavior 2ry to other disorders.

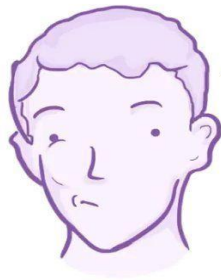
risperidone and *aripiprazole* are approved for secondary to autism

4. Chlorpromazine is used to treat intractable hiccups.

5. Risperidone and haloperidol are also commonly prescribed for tic disorder.

TIC DISORDERS

REPETITIVE
NON-VOLUNTARY
MOVEMENTS
OR
VOCALIZATIONS



BEFORE 18 years old

SIMPLE
e.g. eye blink

COMPLEX
e.g. distracting
gestures

6. Aripiprazole & Quetiapine with antidepressant agents for treatment of refractory depression.

7. Off-label use in treatment of chronic insomnia (small dose of Quetiapine)

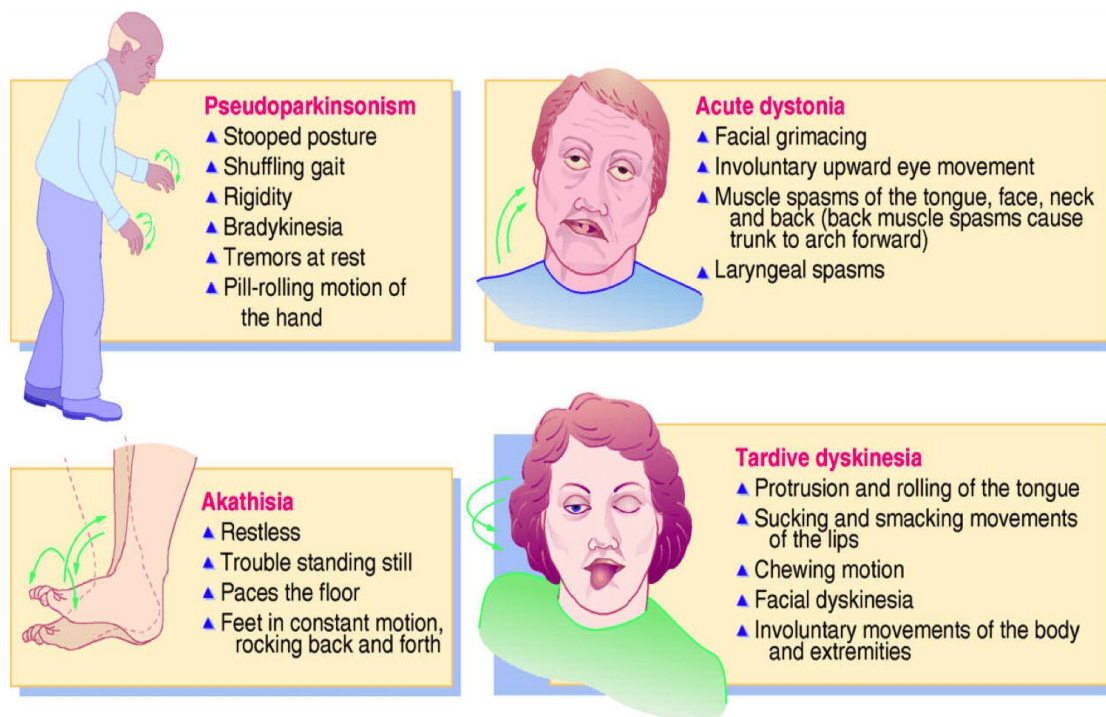
D. Pharmacokinetics

Most of the antipsychotics are very lipophilic and cross lipid membranes freely. When administered orally, they are well absorbed (bioavailability: 10-70%), highly bound to plasma proteins (75-99%) and tissues, and are extensively distributed (VD: 100-1000 L). Primary route of elimination for most of antipsychotics is hepatic metabolism and biotransformation produces active metabolites.

E. Adverse effects

1. Extrapyramidal effects (see the figure):

• The



appearance of the movement disorders is generally time and dose dependent, **dystonia** occurs within a few hours to days of treatment, followed by **akathisia** occurring within days to weeks.

- **Parkinson like symptoms** of bradykinesia, rigidity, and tremor occur within weeks to months.
- **Tardive dyskinesia:** It is a late onset type of extrapyramidal symptoms usually after several months or even years. Manifested as sudden uncontrollable movements happen in the face and the body because of prolonged use of medication, typically antipsychotic drugs → ↑ NO of D receptors as a compensatory response to long-term blockade → neuron supersensitive to actions of dopamine, → excess movement in the patient.
- **Thioridazine** show fewer EPS, because strong anticholinergic
- **haloperidol** show higher EPS, because low anticholinergic

N.B: Extrapyramidal effects are minimized by administration of an anticholinergic drugs (e.g. benztropine).

Akathisia may respond better to β blockers or benzodiazepines, rather than anticholinergic.

Tardive dyskinesia is treated with Valbenazine: it is thought to function as a highly selective inhibitor of the VMAT2 vesicular monoamine transporter resulting in decreased availability of dopamine in the presynaptic cleft. This leads to decreased dopaminergic activation of the striatal motor pathway.

2. Neuroleptic malignant syndrome:

Fatal reaction characterized by muscle rigidity, fever, altered mental status and stupor, unstable blood pressure, and myoglobinemia.

Treatment necessitates discontinuation of the antipsychotic & supportive therapy.

Administration of dantrolene and bromocriptine are helpful.

4. Other effects:

- ❖ Drowsiness during the first few weeks of treatment (due to CNS depression & antihistaminic).
- ❖ Agents with potent antimuscarinic activity → dry mouth, urinary retention, constipation, and loss of visual accommodation (atropine like action).
- ❖ Others block α -adrenergic receptors → ↓ BP & orthostatic hypotension.
- ❖ Depression of hypothalamus, affecting thermoregulation.
- ❖ Hyperprolactinemia → amenorrhea, galactorrhea, gynecomastia, infertility, and erectile dysfunction.
- ❖ Metabolic effects: risperidone may cause insulin resistance and induce DM.
- ❖ Bone marrow suppression specially with clozapine.

Refractory patients:

10% to 20% of patients with schizophrenia have an insufficient response

Clozapine effective antipsychotic with a minimal risk of EPS, its clinical use is limited to refractory patients because of bone marrow suppression, seizures, and cardiovascular side effects, as orthostasis.

Severe agranulocytosis necessitates frequent monitoring of white blood cell counts.

Maintenance treatment

Patients have had two or more psychotic episodes 2ry to schizophrenia

→maintenance therapy for at least 5 years.